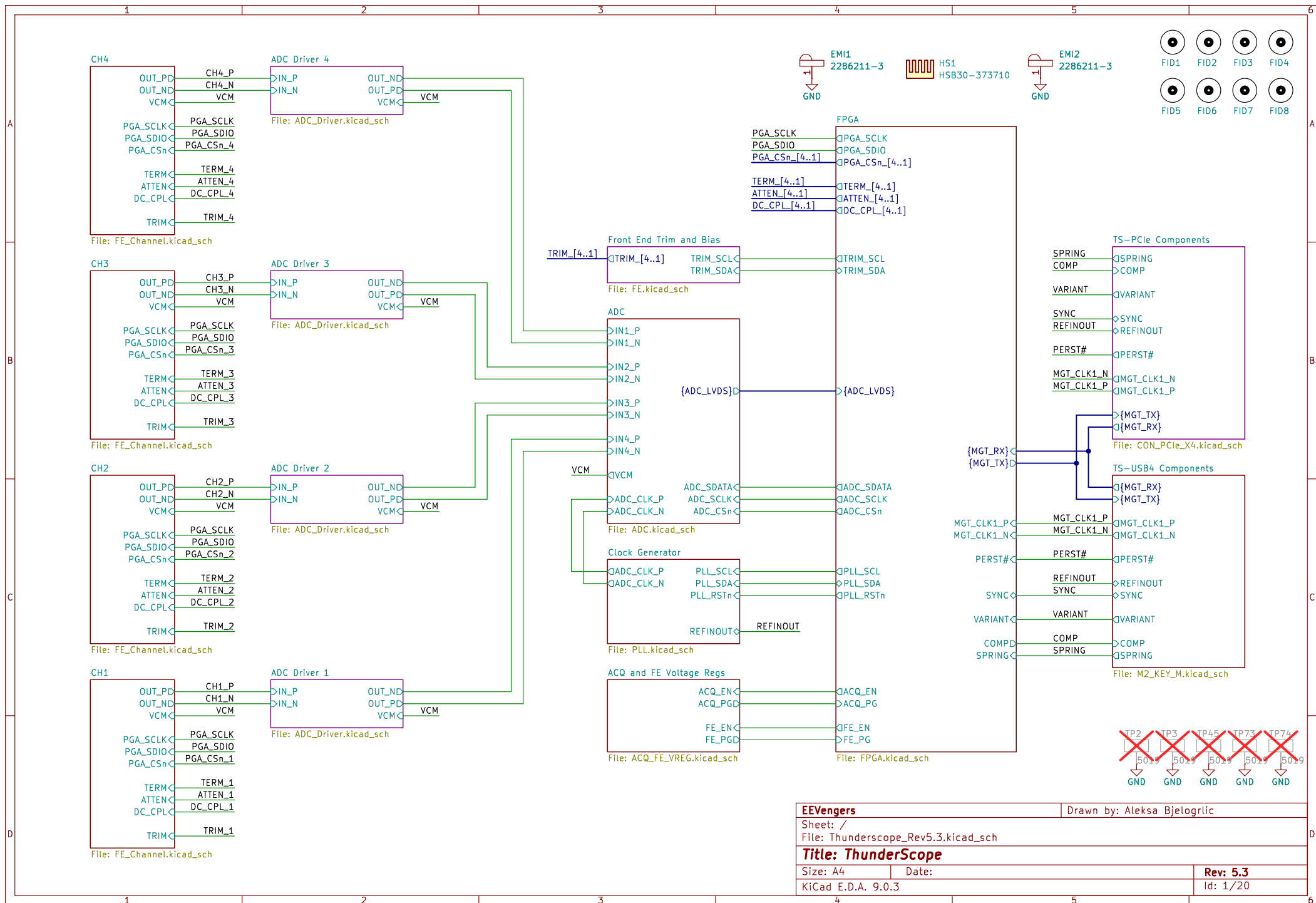
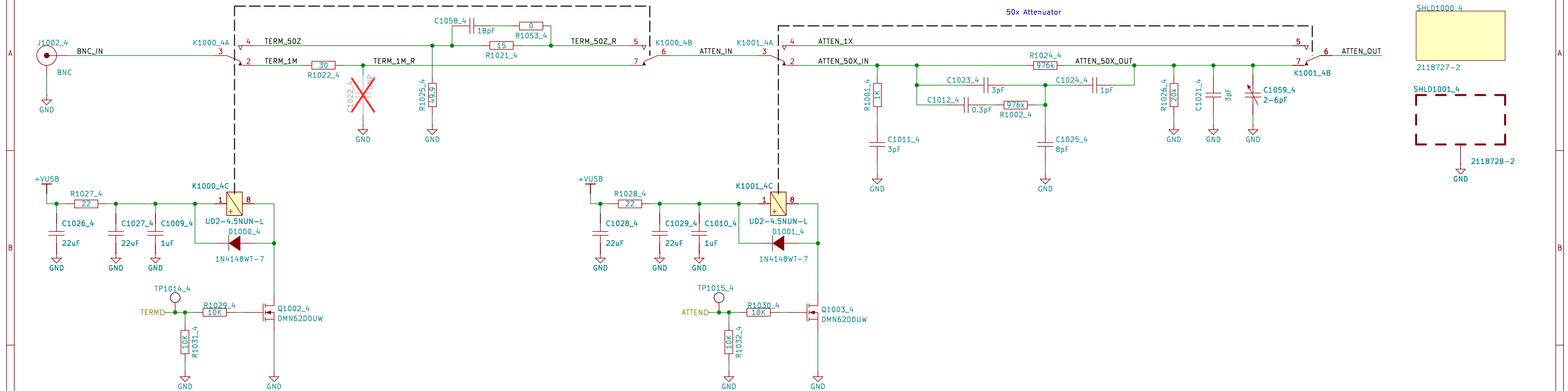
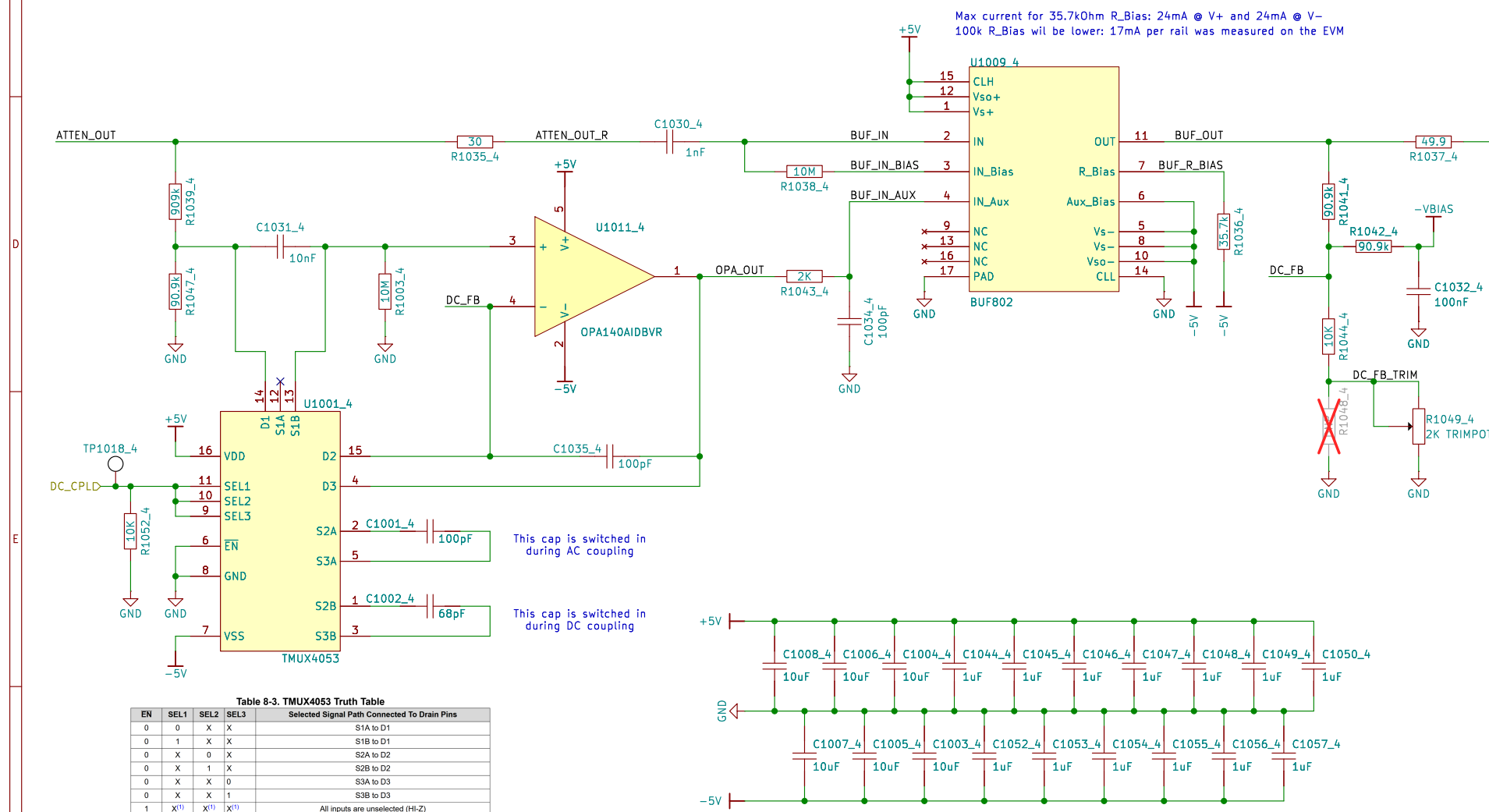




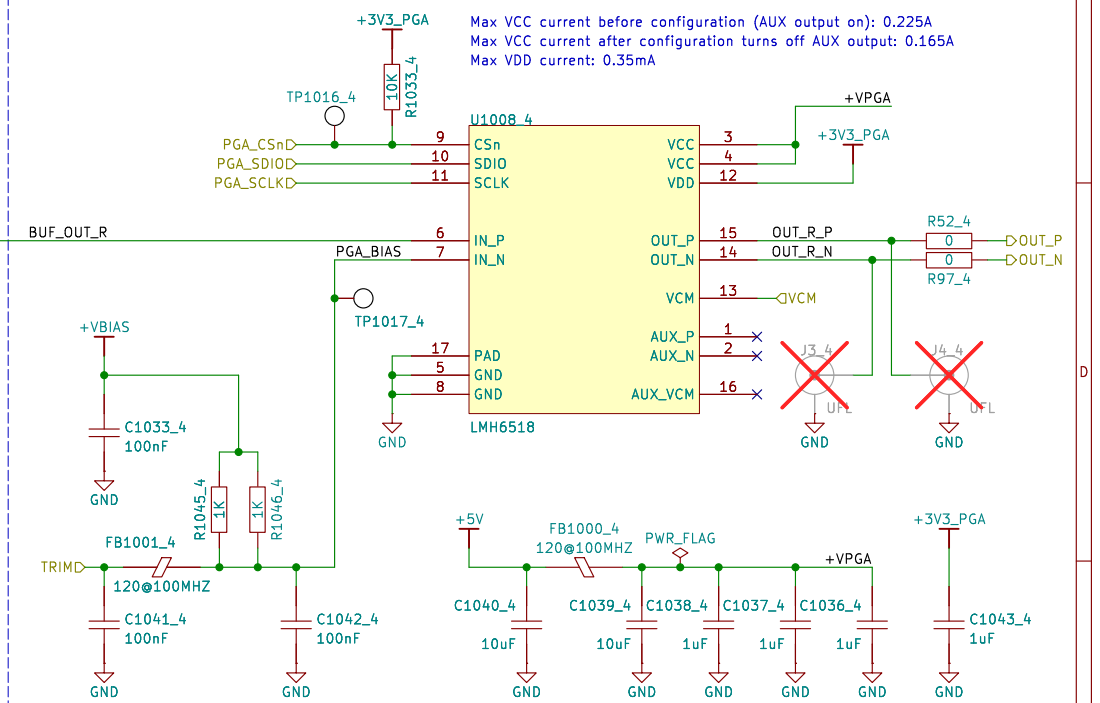
## Termination and Attenuation



## Input Buffer and AC/DC Coupling

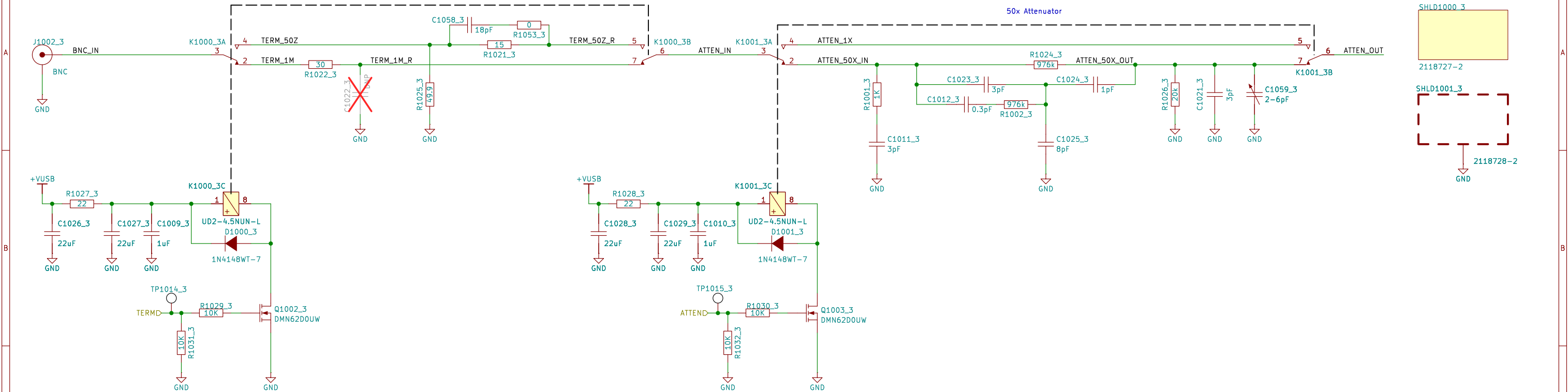


## Programmable Gain Amplifier

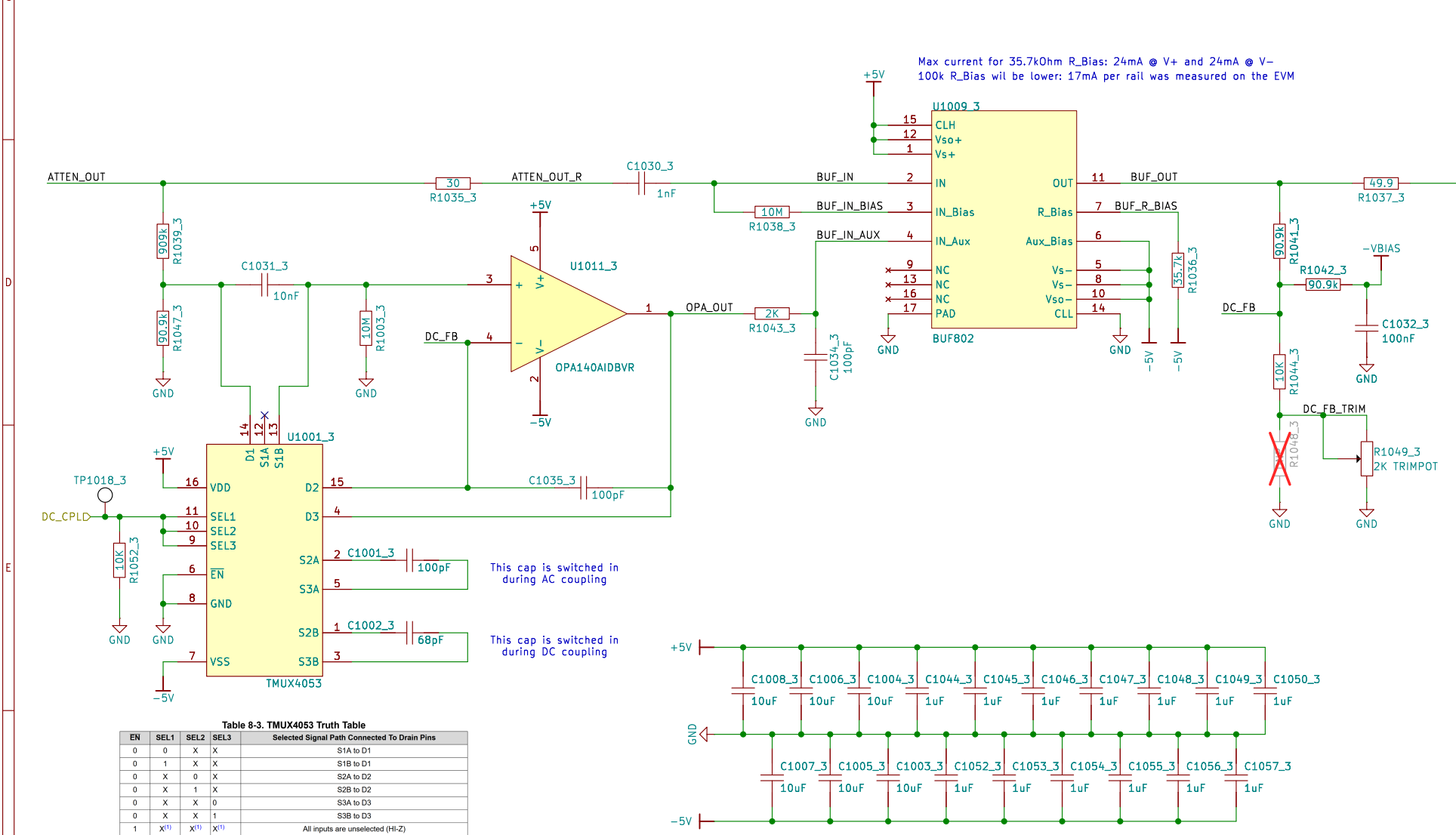


| EN | SEL1             | SEL2             | SEL3             | Selected Signal Path Connected To Drain Pins |
|----|------------------|------------------|------------------|----------------------------------------------|
| 0  | 0                | X                | X                | S1A to D1                                    |
| 0  | 1                | X                | X                | S1B to D1                                    |
| 0  | X                | 0                | X                | S2A to D2                                    |
| 0  | X                | 1                | X                | S2B to D2                                    |
| 0  | X                | X                | 0                | S3A to D3                                    |
| 0  | X                | X                | 1                | S3B to D3                                    |
| 1  | X <sup>(1)</sup> | X <sup>(1)</sup> | X <sup>(1)</sup> | All inputs are unselected (Hi-Z)             |

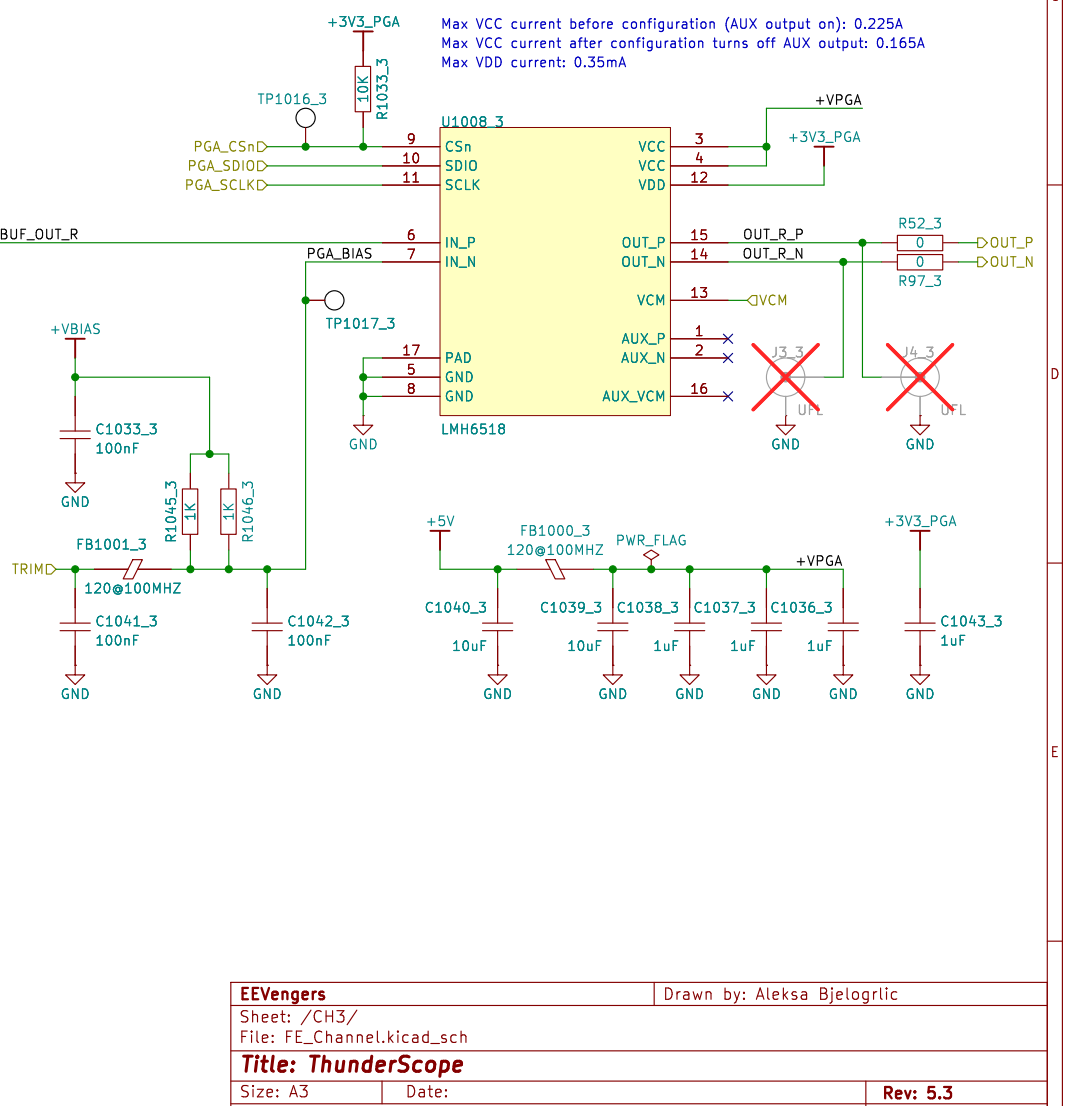
Termination and Attenuation



Input Buffer and AC/DC Coupling



Programmable Gain Amplifier



The PCB layout for the Termination and Attenuation circuit is shown. It includes a BNC input (J1002\_2) connected to a termination network (K1000\_2A) with resistors R1022\_2, R1025\_2, and R1053\_2. A 50x Attenuator (K1001\_2A) is also present, with resistors R1024\_2, R1025\_2, and R1026\_2. The circuit is powered by +VUSB and includes a diode UD2-4.5NUN-L (D1000\_2) and a MOSFET Q1002\_2 (DMN62D0UW) for signal processing. The output is connected to a 50Ω load (K1000\_2B) and a 50x Attenuator (K1001\_2B) with resistors R1024\_2, R1025\_2, and R1026\_2. The layout is labeled with component values and reference designators.

Max current for 35.7kOhm R<sub>Bias</sub>: 24mA @ V<sub>+</sub> and 24mA @ V<sub>-</sub>  
100k R<sub>Bias</sub> will be lower: 17mA per rail was measured on the EVM

Table 8-3. TMUX4053 Truth Table

| EN | SEL1             | SEL2             | SEL3             | Selected Signal Path Connected to Drain Pins |
|----|------------------|------------------|------------------|----------------------------------------------|
| 0  | 0                | X                | X                | S1A to D1                                    |
| 0  | 1                | X                | X                | S1B to D1                                    |
| 0  | X                | 0                | X                | S2A to D2                                    |
| 0  | X                | 1                | X                | S2B to D2                                    |
| 0  | X                | X                | 0                | S3A to D3                                    |
| 0  | X                | X                | 1                | S3B to D3                                    |
| 1  | X <sup>(1)</sup> | X <sup>(1)</sup> | X <sup>(1)</sup> | All inputs are unselected (Hi-Z)             |

**Programmable Gain Amplifier**

Max VCC current before configuration (AUX output on): 0.225A  
 Max VCC current after configuration turns off AUX output: 0.165A  
 Max VDD current: 0.35mA

U1008\_2

CSn 9  
 SDIO 10  
 SCLK 11

VCC 3  
 VCC 4  
 VDD 12

IN\_P 6  
 IN\_N 7

OUT\_P 15  
 OUT\_N 14

VCM 13

AUX\_P 1  
 AUX\_N 2

AUX\_VCM 16

LMH6518

TP1016\_2

TP1017\_2

TP1018\_2

TP1019\_2

TP1020\_2

TP1021\_2

TP1022\_2

TP1023\_2

TP1024\_2

TP1025\_2

TP1026\_2

TP1027\_2

TP1028\_2

TP1029\_2

TP1030\_2

TP1031\_2

TP1032\_2

TP1033\_2

TP1034\_2

TP1035\_2

TP1036\_2

TP1037\_2

TP1038\_2

TP1039\_2

TP1040\_2

TP1041\_2

TP1042\_2

TP1043\_2

TP1044\_2

TP1045\_2

TP1046\_2

TP1047\_2

TP1048\_2

TP1049\_2

TP1050\_2

TP1051\_2

TP1052\_2

TP1053\_2

TP1054\_2

TP1055\_2

TP1056\_2

TP1057\_2

TP1058\_2

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TP1070\_2

TP1071\_2

TP1072\_2

TP1073\_2

TP1074\_2

TP1075\_2

TP1076\_2

TP1077\_2

TP1078\_2

TP1079\_2

TP1080\_2

TP1081\_2

TP1082\_2

TP1083\_2

TP1084\_2

TP1085\_2

TP1086\_2

TP1087\_2

TP1088\_2

TP1089\_2

TP1090\_2

TP1091\_2

TP1092\_2

TP1093\_2

TP1094\_2

TP1095\_2

TP1096\_2

TP1097\_2

TP1098\_2

TP1099\_2

TP1100\_2

TP1101\_2

TP1102\_2

TP1103\_2

TP1104\_2

TP1105\_2

TP1106\_2

TP1107\_2

TP1108\_2

TP1109\_2

TP1110\_2

TP1111\_2

TP1112\_2

TP1113\_2

TP1114\_2

TP1115\_2

TP1116\_2

TP1117\_2

TP1118\_2

TP1119\_2

TP1120\_2

TP1121\_2

TP1122\_2

TP1123\_2

TP1124\_2

TP1125\_2

TP1126\_2

TP1127\_2

TP1128\_2

TP1129\_2

TP1130\_2

TP1131\_2

TP1132\_2

TP1133\_2

TP1134\_2

TP1135\_2

TP1136\_2

TP1137\_2

TP1138\_2

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TP1174\_2

TP1175\_2

TP1176\_2

TP1177\_2

TP1178\_2

TP1179\_2

TP1180\_2

TP1181\_2

TP1182\_2

TP1183\_2

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TP1185\_2

TP1186\_2

TP1187\_2

TP1188\_2

TP1189\_2

TP1190\_2

TP1191\_2

TP1192\_2

TP1193\_2

TP1194\_2

TP1195\_2

TP1196\_2

TP1197\_2

TP1198\_2

TP1199\_2

TP1200\_2

TP1201\_2

TP1202\_2

TP1203\_2

TP1204\_2

TP1205\_2

TP1206\_2

TP1207\_2

TP1208\_2

TP1209\_2

TP1210\_2

TP1211\_2

TP1212\_2

TP1213\_2

TP1214\_2

TP1215\_2

TP1216\_2

TP1217\_2

TP1218\_2

TP1219\_2

TP1220\_2

TP1221\_2

TP1222\_2

TP1223\_2

TP1224\_2

TP1225\_2

TP1226\_2

TP1227\_2

TP1228\_2

TP1229\_2

TP1230\_2

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TP1232\_2

TP1233\_2

TP1234\_2

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TP1250\_2

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TP1256\_2

TP1257\_2

TP1258\_2

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TP1262\_2

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TP1265\_2

TP1266\_2

TP1267\_2

TP1268\_2

TP1269\_2

TP1270\_2

TP1271\_2

TP1272\_2

TP1273\_2

TP1274\_2

TP1275\_2

TP1276\_2

TP1277\_2

TP1278\_2

TP1279\_2

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TP1283\_2

TP1284\_2

TP1285\_2

TP1286\_2

TP1287\_2

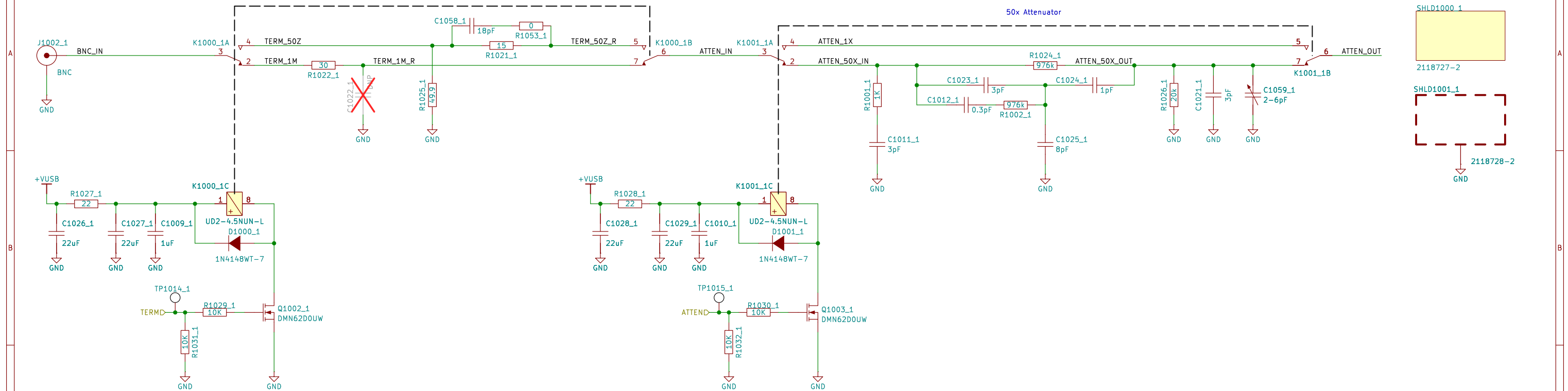
TP1288\_2

TP1289\_2

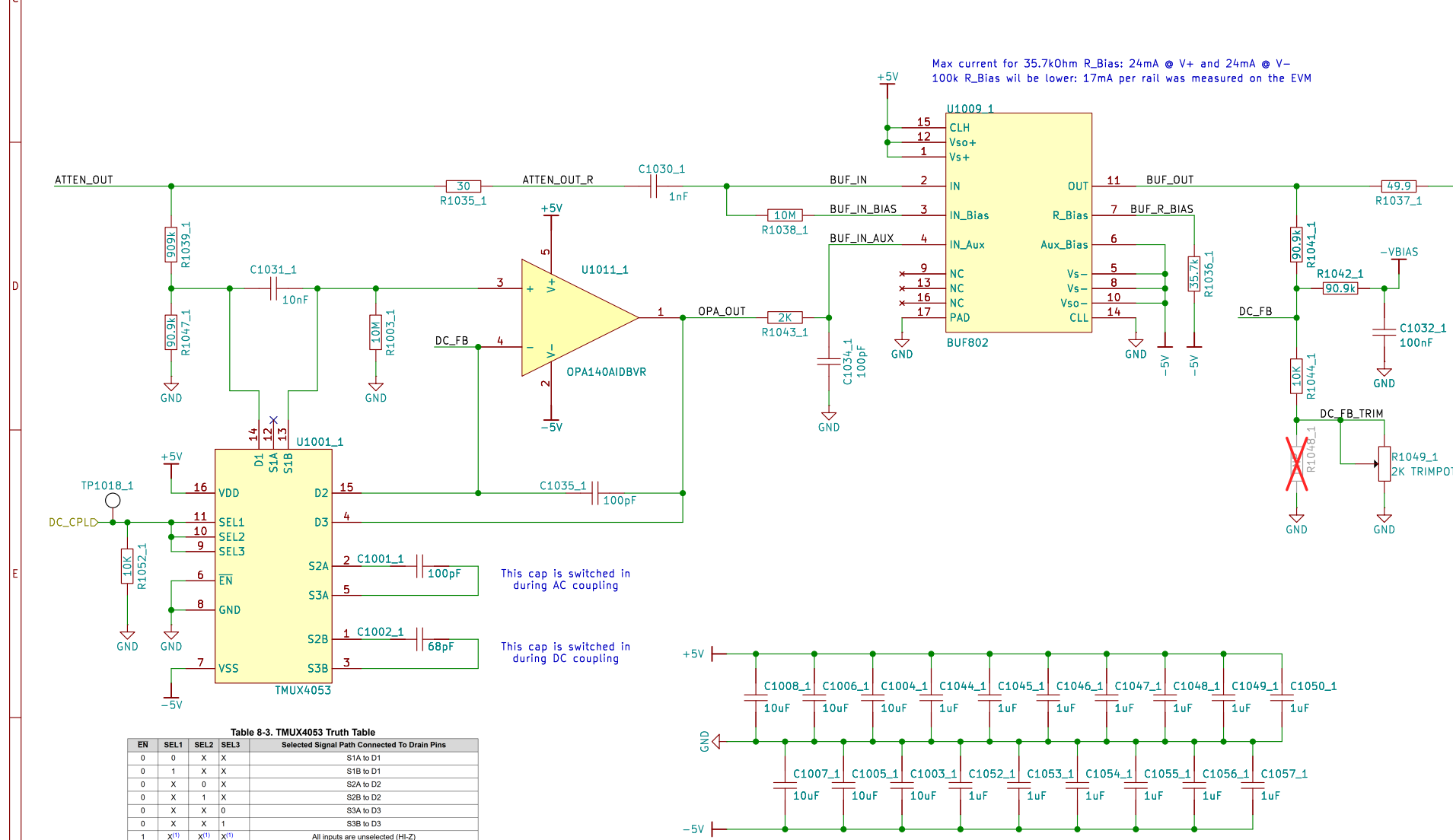
TP1290\_2

TP1291\_2

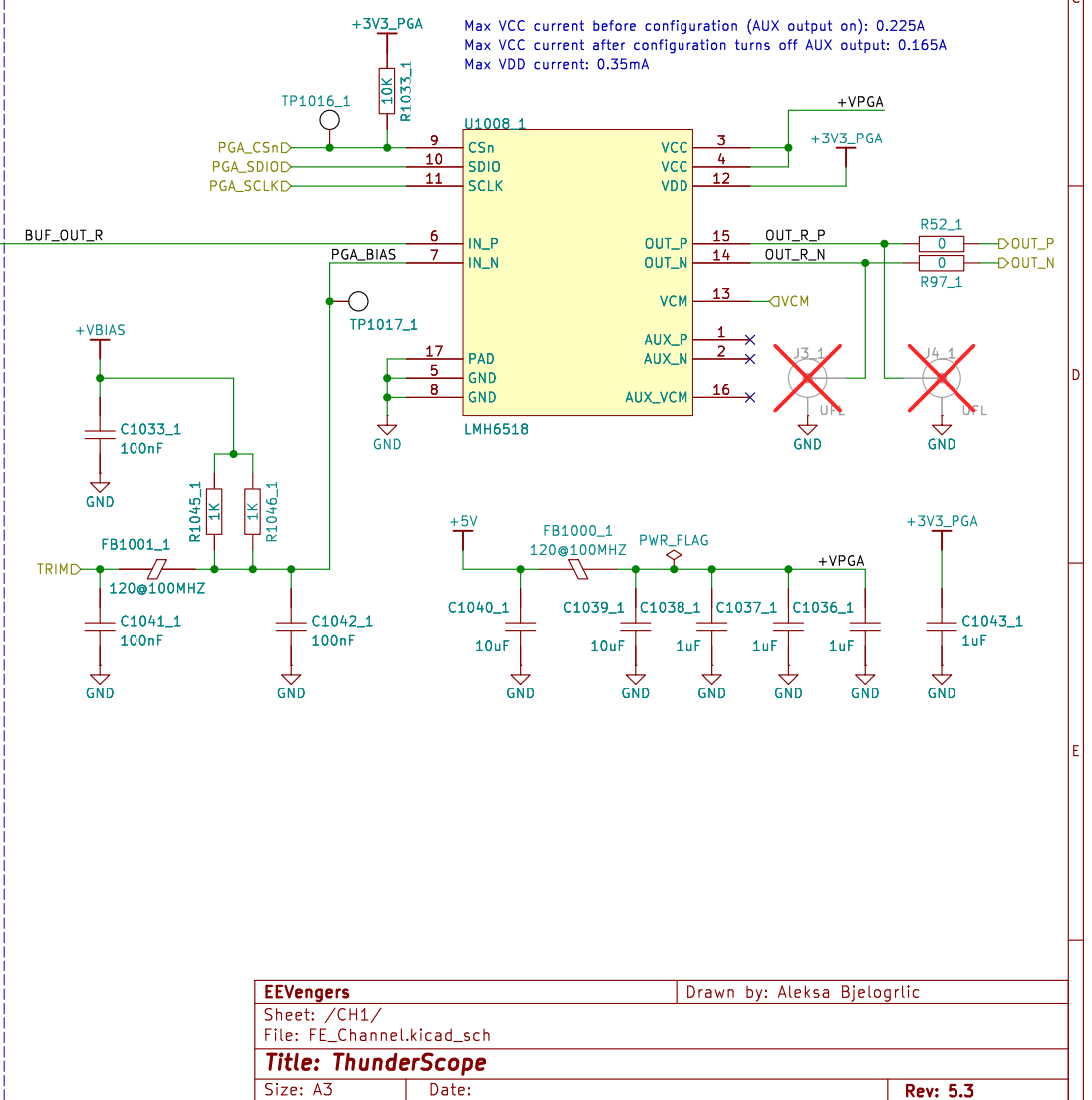
## Termination and Attenuation



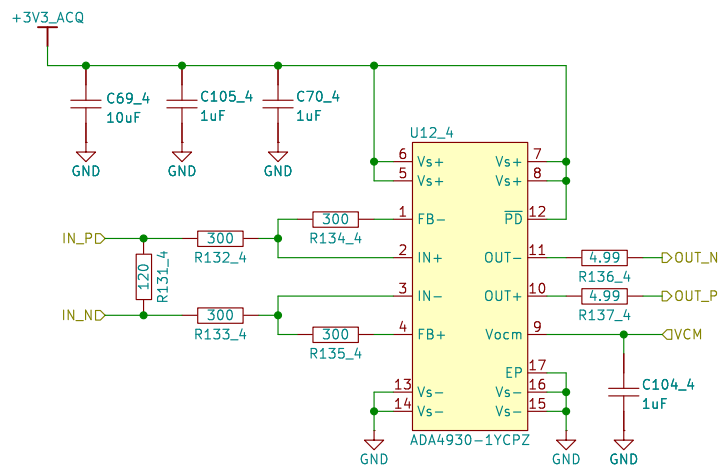
## Input Buffer and AC/DC Coupling



## Programmable Gain Amplifier



# ADC Driver

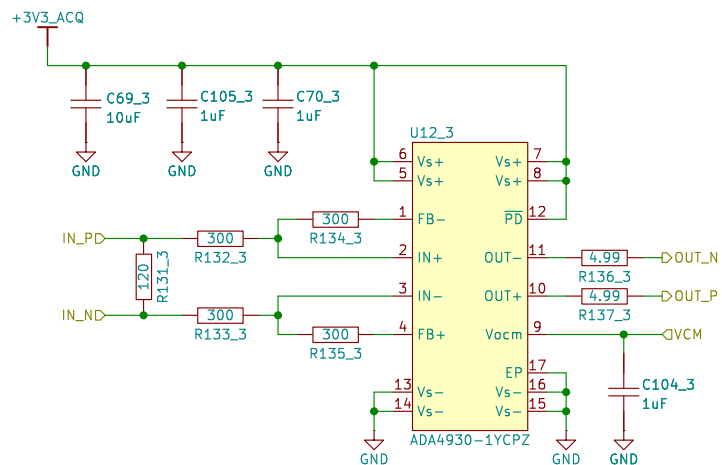


|                                                     |  |                            |  |
|-----------------------------------------------------|--|----------------------------|--|
| EEVengers                                           |  | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /ADC Driver 4/<br>File: ADC_Driver.kicad_sch |  |                            |  |
| Title: ThunderScope                                 |  |                            |  |
| Size: A4                                            |  | Date:                      |  |
| KiCad E.D.A. 9.0.3                                  |  | Rev: 5.3<br>Id: 3/20       |  |

**ADC Driver**

The schematic diagram illustrates an ADC Driver circuit centered around the ADA4930-1YCPZ operational amplifier (U12\_3). The circuit is powered by a +3V3\_ACQ supply, which is decoupled by three capacitors: C69\_3 (10uF), C105\_3 (1uF), and C70\_3 (1uF). The op-amp's non-inverting inputs (IN+) and inverting inputs (IN-) are connected to the IN\_PD and IN\_ND signals through a network of resistors (R131\_3, R132\_3, R133\_3, R134\_3, R135\_3). The feedback network consists of resistors R136\_3 and R137\_3, which are connected to the output pins (OUT- and OUT+) and the common-mode voltage pin (Vcm). The output signals are OUT\_N and OUT\_P. The op-amp's power supply pins (Vs+, Vs-, and Vcm) are connected to the +3V3\_ACQ supply and ground (GND). The Vcm pin is also connected to a common-mode voltage source (VCM) through a 4.99 ohm resistor. The op-amp is configured as a differential amplifier with a gain of 1.

|                                                     |       |                              |
|-----------------------------------------------------|-------|------------------------------|
| <b>EEVengers</b>                                    |       | Drawn by: Aleksa Bjelogrljic |
| Sheet: /ADC Driver 3/<br>File: ADC_Driver.kicad_sch |       |                              |
| <b>Title: ThunderScope</b>                          |       |                              |
| Size: A4                                            | Date: | <b>Rev: 5.3</b>              |
| KiCad E.D.A. 9.0.3                                  |       | Id: 3/20                     |



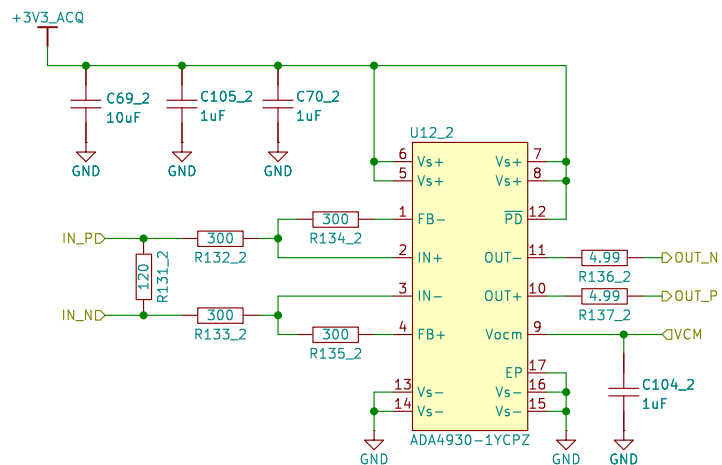
|                                                     |       |                            |
|-----------------------------------------------------|-------|----------------------------|
| <b>EEVengers</b>                                    |       | Drawn by: Aleksa Bjelogrić |
| Sheet: /ADC Driver 3/<br>File: ADC_Driver.kicad_sch |       |                            |
| <b>Title: ThunderScope</b>                          |       |                            |
| Size: A4                                            | Date: | <b>Rev: 5.3</b>            |
| KiCad E.D.A. 9.0.3                                  |       | Id: 3/20                   |

**EEVengers** | Drawn by: Aleksa Bjelogrić

Sheet: /ADC Driver 2/  
File: ADC\_Driver.kicad\_sch

**Title: ThunderScope**

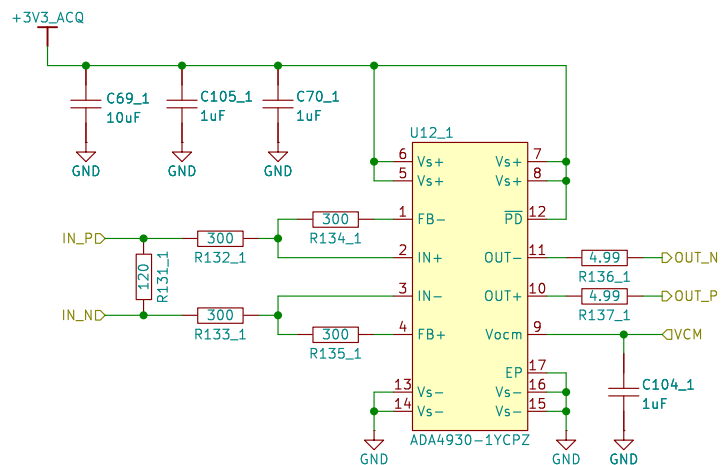
|                    |       |                 |
|--------------------|-------|-----------------|
| Size: A4           | Date: | <b>Rev: 5.3</b> |
| KiCad E.D.A. 9.0.3 |       | Id: 3/20        |



|                                                     |       |                            |  |
|-----------------------------------------------------|-------|----------------------------|--|
| <b>EEVengers</b>                                    |       | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /ADC Driver 2/<br>File: ADC_Driver.kicad_sch |       |                            |  |
| <b>Title: ThunderScope</b>                          |       |                            |  |
| Size: A4                                            | Date: | Rev: <b>5.3</b>            |  |
| KiCad E.D.A. 9.0.3                                  |       | Id: 3/20                   |  |

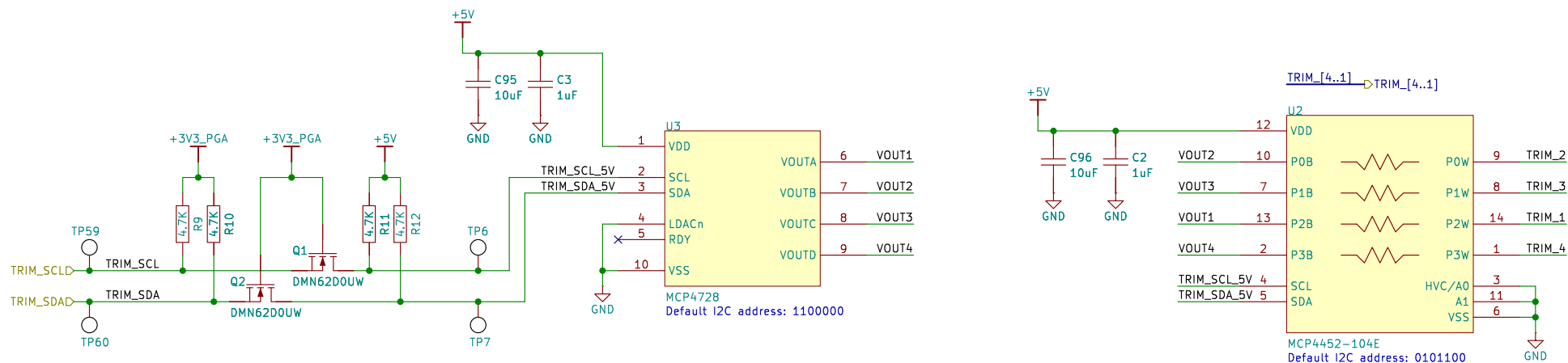


# ADC Driver

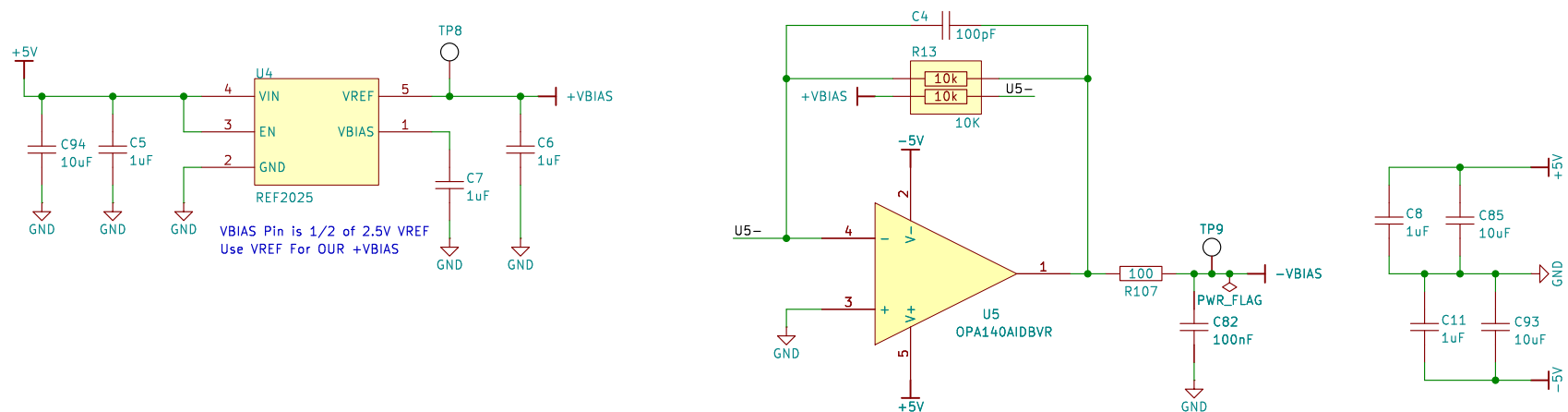


|                                                     |  |                            |  |
|-----------------------------------------------------|--|----------------------------|--|
| EEVengers                                           |  | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /ADC Driver 1/<br>File: ADC_Driver.kicad_sch |  |                            |  |
| Title: ThunderScope                                 |  |                            |  |
| Size: A4                                            |  | Date:                      |  |
| KiCad E.D.A. 9.0.3                                  |  | Rev: 5.3<br>Id: 3/20       |  |

## Offset Voltage Trim and User Offset Control

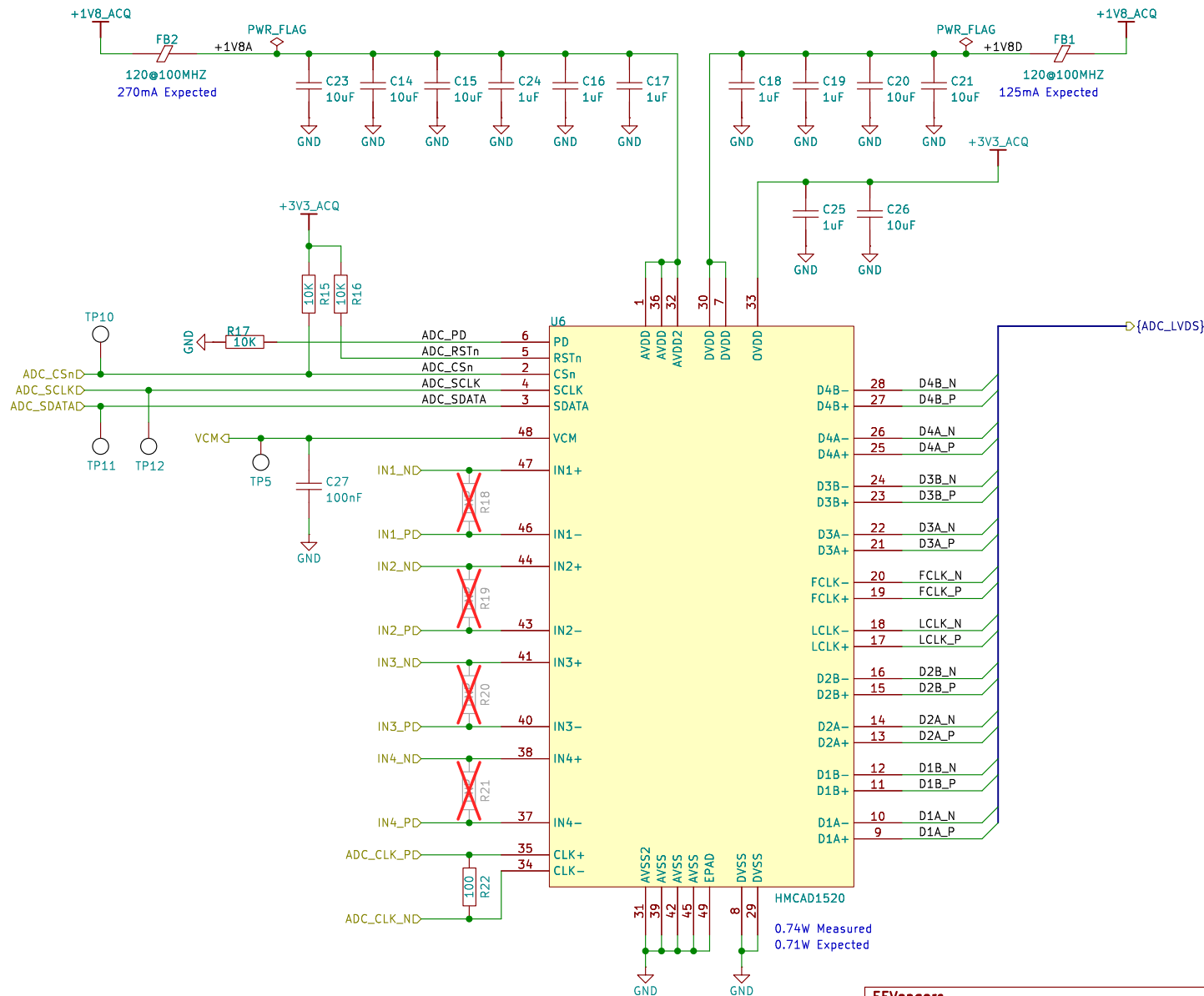


## Bias Voltage Generation



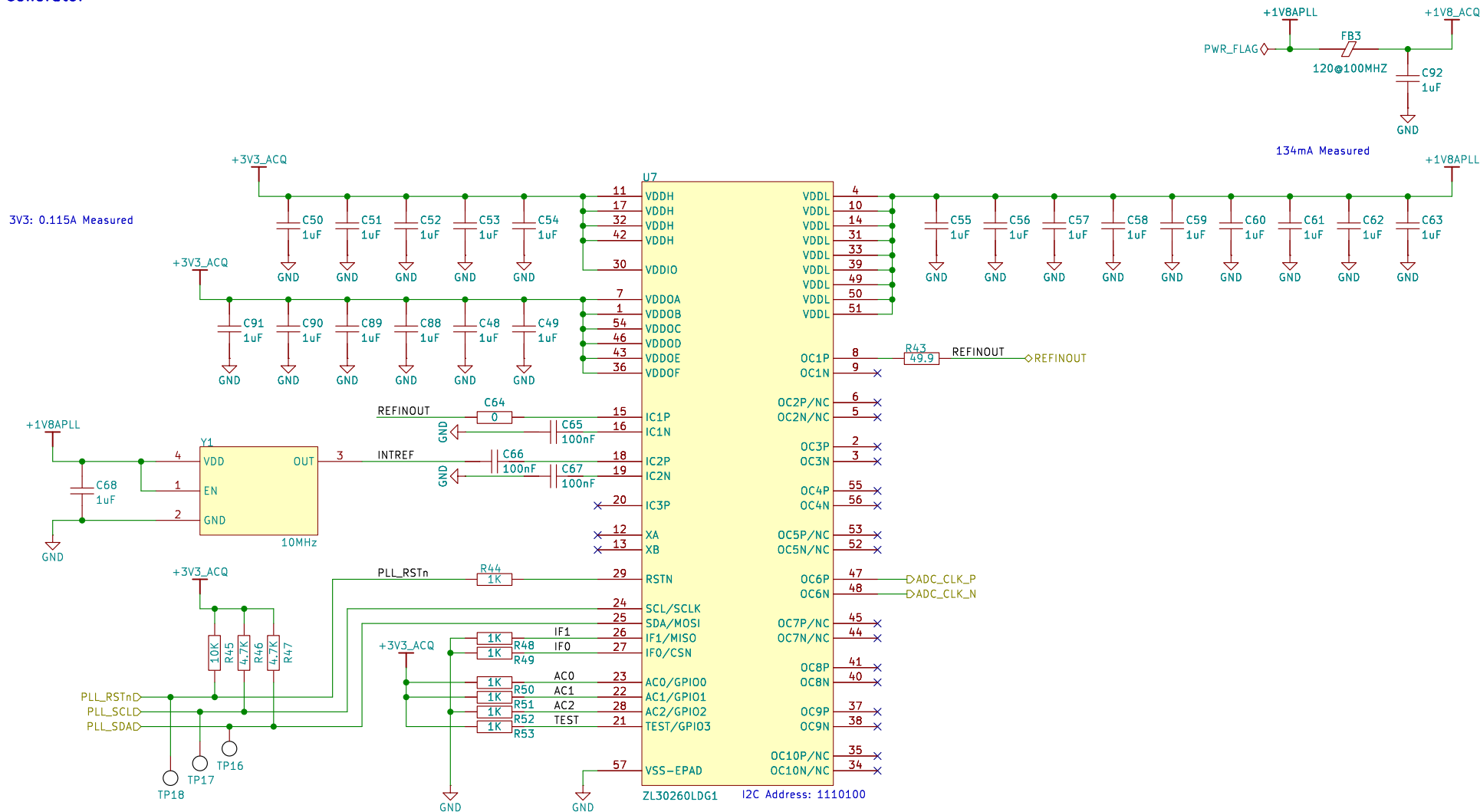
|                                                        |       |                              |          |
|--------------------------------------------------------|-------|------------------------------|----------|
| EEVengers                                              |       | Drawn by: Aleksa Bjelogrltic |          |
| Sheet: /Front End Trim and Bias/<br>File: FE.kicad_sch |       |                              |          |
| Title: ThunderScope                                    |       |                              |          |
| Size: A4                                               | Date: |                              | Rev: 5.3 |
| KiCad E.D.A. 9.0.3                                     |       |                              | Id: 4/20 |

# ADC



|                     |       |                            |          |
|---------------------|-------|----------------------------|----------|
| EEVengers           |       | Drawn by: Aleksa Bjelogrlc |          |
| Sheet: /ADC/        |       |                            |          |
| File: ADC.kicad_sch |       |                            |          |
| Title: ThunderScope |       |                            |          |
| Size: A4            | Date: |                            | Rev: 5.3 |
| KiCad E.D.A. 9.0.3  |       |                            | Id: 5/20 |

## Clock Generator



| IF1 | IF0 | Processor Interface                                 | Configuration Memory to Use |
|-----|-----|-----------------------------------------------------|-----------------------------|
| 0   | 0   | I <sup>2</sup> C, slave address 11101 00            | Internal ROM                |
| 0   | 1   | I <sup>2</sup> C, slave address 11101 01            | Internal ROM                |
| 1   | 0   | SPI Slave                                           | Internal ROM                |
| 1   | 1   | SPI Master during auto-configuration then SPI Slave | External SPI EEPROM         |

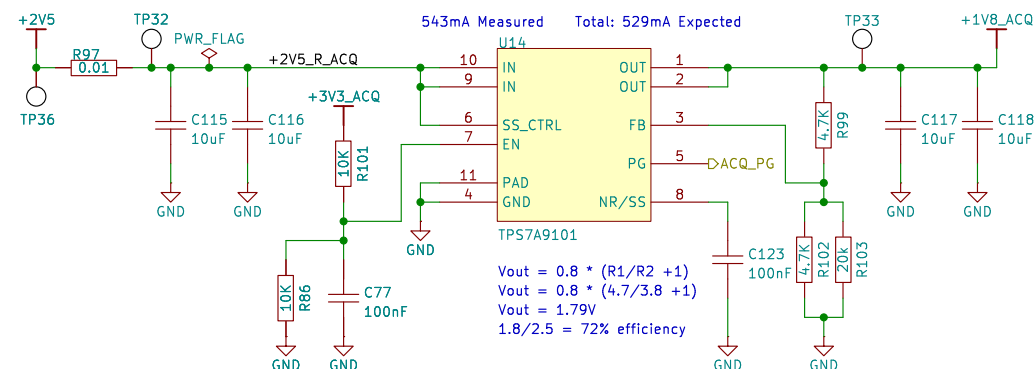
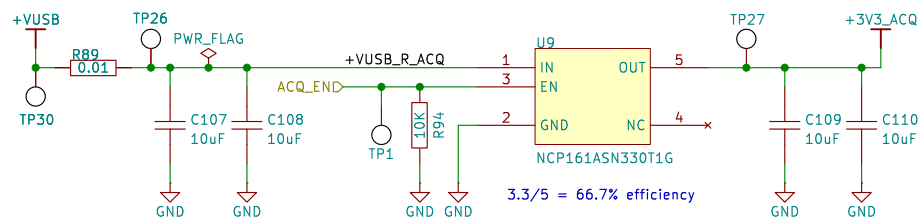
To configure the device as specified in the first three rows above but without auto-configuring from internal ROM, wire devices pins as follows: TEST=1 and AC[2:0]=011, as described in section 5.2.

| AC2 | AC1 | AC0 | Auto Configuration |
|-----|-----|-----|--------------------|
| 0   | 0   | 0   | Configuration 0    |
| 0   | 0   | 1   | Configuration 1    |
| 0   | 1   | 0   | Configuration 2    |
| 0   | 1   | 1   | Configuration 3    |
| 1   | 0   | 0   | Configuration 4    |
| 1   | 0   | 1   | Configuration 5    |
| 1   | 1   | 0   | Configuration 6    |
| 1   | 1   | 1   | Configuration 7    |

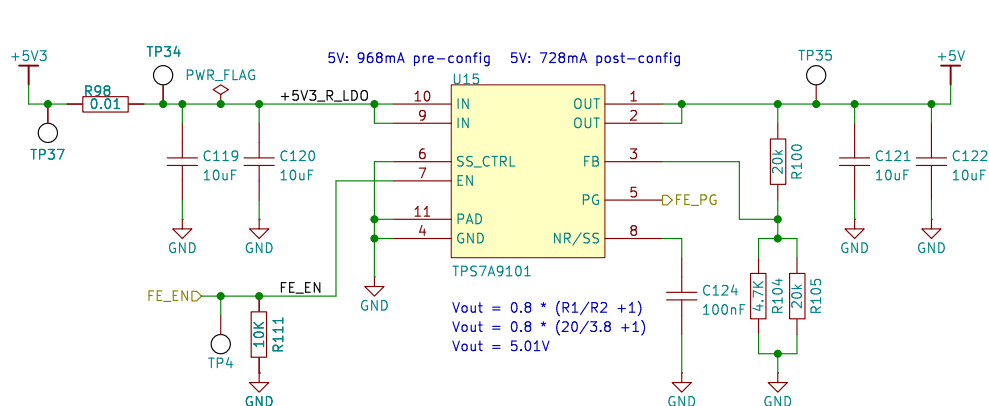
3.3 + 1.8V operation w/ one input one output: 0.67W Expected  
0.62W Measured

|                                                 |       |                            |          |
|-------------------------------------------------|-------|----------------------------|----------|
| EEVengers                                       |       | Drawn by: Aleksa Bjelogrić |          |
| Sheet: /Clock Generator/<br>File: PLL.kicad_sch |       |                            |          |
| Title: <b>ThunderScope</b>                      |       |                            |          |
| Size: A4                                        | Date: |                            | Rev: 5.3 |
| KiCad E.D.A. 9.0.3                              |       |                            | Id: 6/20 |

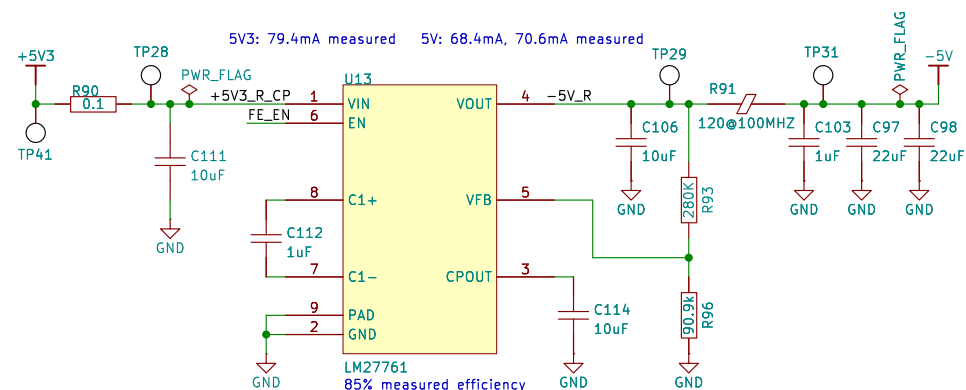
## Acquisition Voltage Regulators



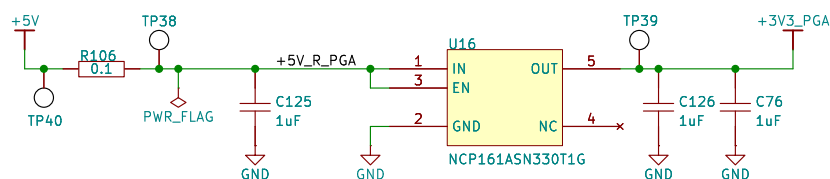
## Front End Voltage Regulators



6ms soft start time on previous LDO, need same or greater

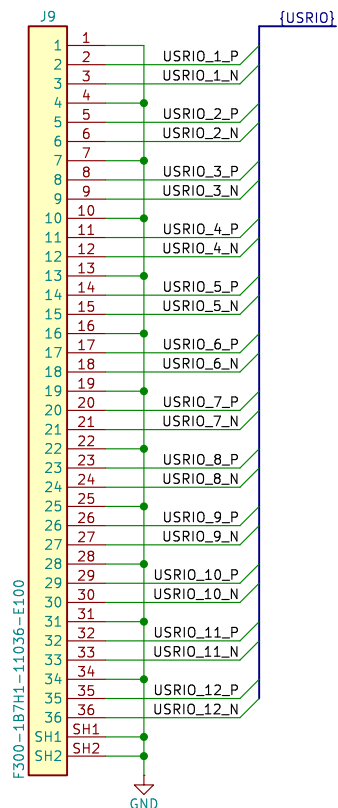
$$T_{SS} = (V_{REF} \times C_{nr/ss}) / I_{nr/ss}$$
$$T_{SS} = (0.8 \times 100nF) / 6.2\mu A \text{ [for SS_CTRL = GND]}$$
$$T_{SS} = 0.0129s = 13ms$$


$V_{out} = -1.22V \cdot (R_1 + R_2) / R_2$   
 $V_{out} = -1.22 \cdot (280 / 90.9 + 1) = -4.98V$   
 The value for  $R_2$  must be no less than 50 kΩ.



|                                                                 |       |                            |  |
|-----------------------------------------------------------------|-------|----------------------------|--|
| <b>EEVengers</b>                                                |       | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /ACQ and FE Voltage Regs/<br>File: ACQ_FE_VREG.kicad_sch |       |                            |  |
| <b>Title: ThunderScope</b>                                      |       |                            |  |
| Size: A4                                                        | Date: | <b>Rev: 5.3</b>            |  |
| KiCad E.D.A. 9.0.3                                              |       | Id: 7/20                   |  |

## User IO Flat Flex Connector



### FPGA Power Inputs



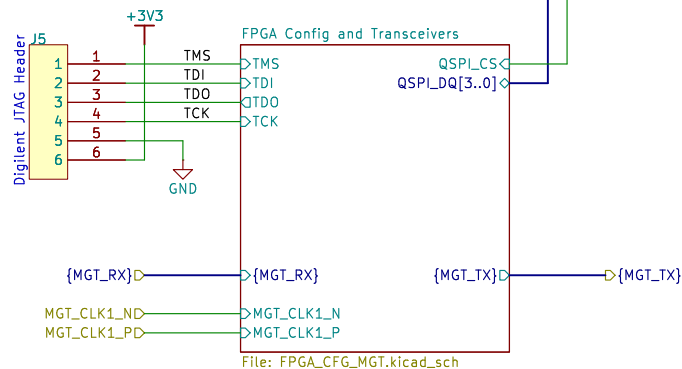
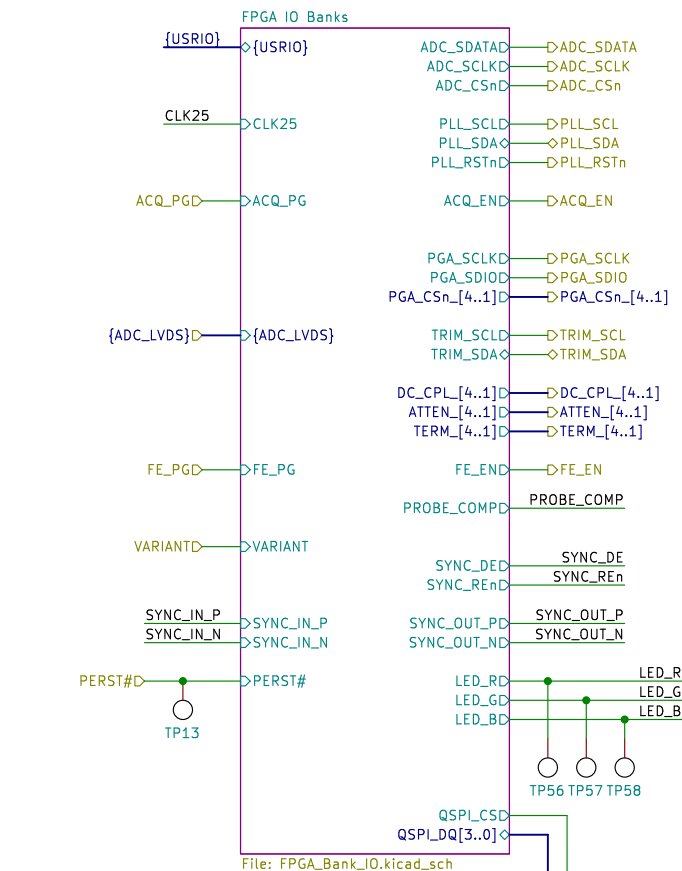
File: FPGA\_PWR.kicad\_sch

### FPGA Voltage Regs

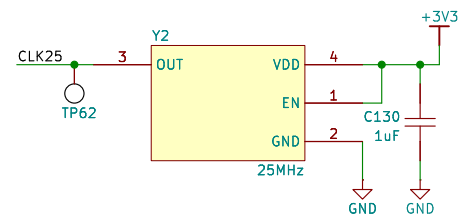


File: FPGA\_VREG.kicad\_sch

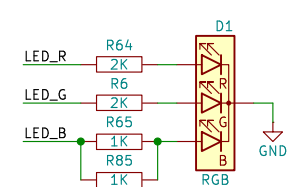
## FPGA



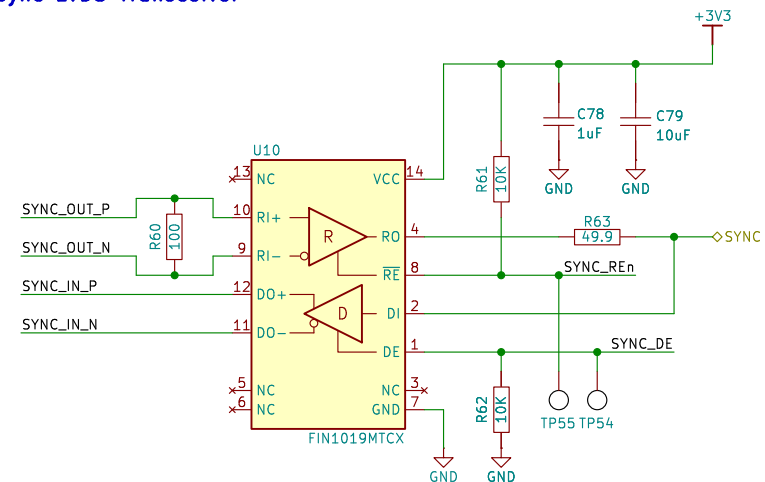
## Always On Clock



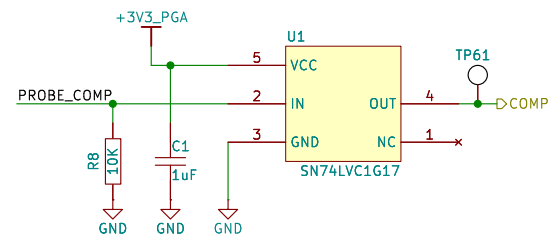
## Status LED



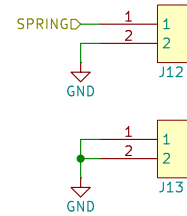
## Sync LVDS Transceiver



## Probe Compensation



## Spring Connectors



EEVengers

Drawn by: Aleksa Bjelogrić

Sheet: /FPGA/

File: FPGA.kicad\_sch

**Title: ThunderScope**

Size: A4

Date:

KiCad E.D.A. 9.0.3

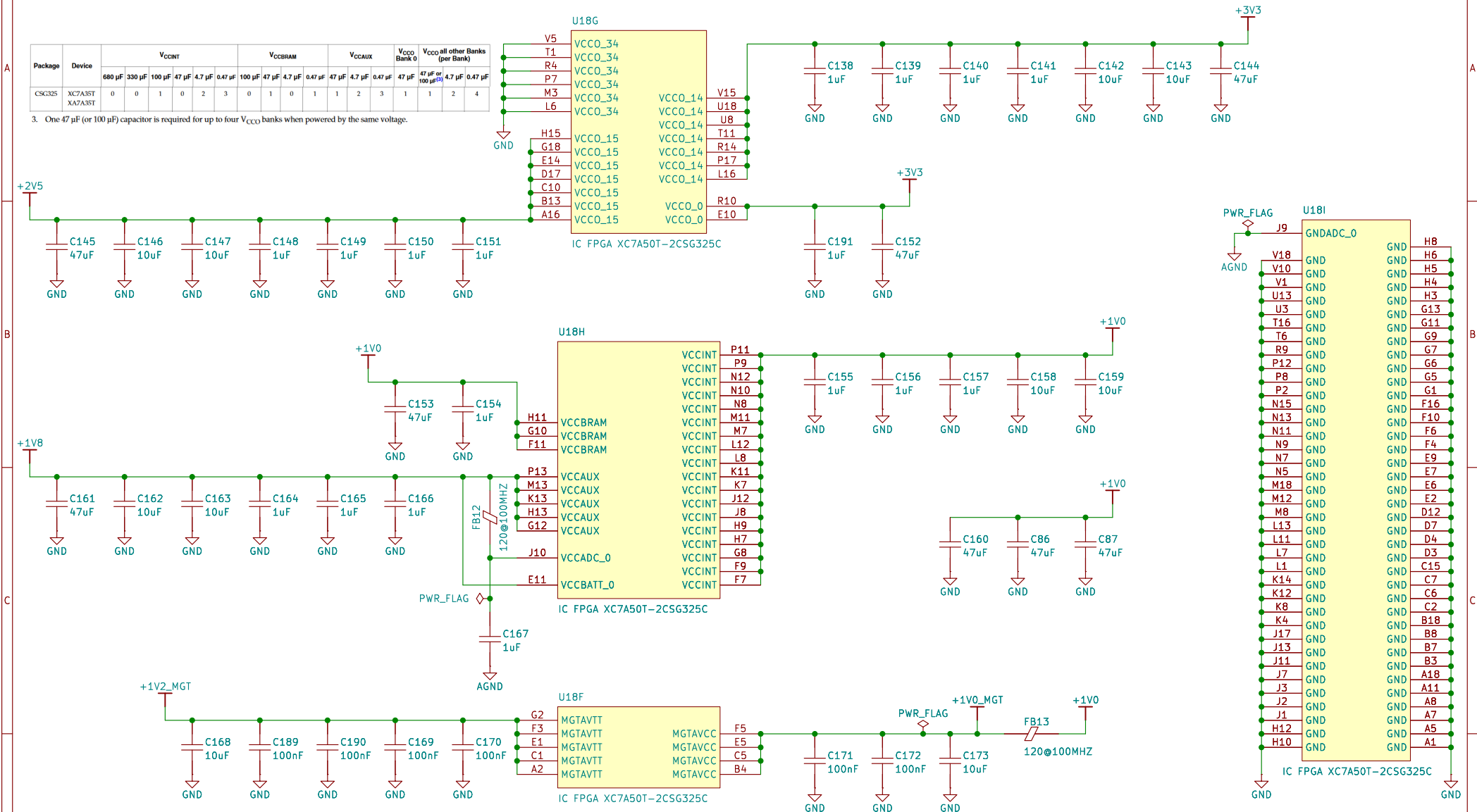
Rev: 5.3

Id: 8/20

# FPGA Power Inputs

| Package | Device             | VCCINT |        |        |       |        |         | VCCBRAM |       |        |         |       |        | VCCAUX  |        |         |       |        |         | VCC0 Bank 0     | VCC0 all other Banks (per Bank) |         |        |         |
|---------|--------------------|--------|--------|--------|-------|--------|---------|---------|-------|--------|---------|-------|--------|---------|--------|---------|-------|--------|---------|-----------------|---------------------------------|---------|--------|---------|
|         |                    | 680 µF | 330 µF | 100 µF | 47 µF | 4.7 µF | 0.47 µF | 100 µF  | 47 µF | 4.7 µF | 0.47 µF | 47 µF | 4.7 µF | 0.47 µF | 4.7 µF | 0.47 µF | 47 µF | 4.7 µF | 0.47 µF | 47 µF or 100 µF | 4.7 µF                          | 0.47 µF | 4.7 µF | 0.47 µF |
| CSG325  | XC7A35T<br>XA7A35T | 0      | 0      | 1      | 0     | 2      | 3       | 0       | 1     | 0      | 1       | 1     | 2      | 3       | 1      | 1       | 1     | 1      | 2       | 4               |                                 |         |        |         |

3. One 47 µF (or 100 µF) capacitor is required for up to four VCC0 banks when powered by the same voltage.



| Capacitor | Package Pins |         |     | Value  |
|-----------|--------------|---------|-----|--------|
|           | MGTAVCC      | MGTAVTT | GND |        |
| Cap1      |              | F3      | F4  | 0.1 µF |
| Cap2      |              | A2      | A1  |        |
| Cap3      | B4           | B3      | B3  |        |
| Cap4      | F5           |         | F6  |        |

| Qty/Power Supply Group |         | Capacitance (µF) | Tolerance | Type    |
|------------------------|---------|------------------|-----------|---------|
| MGTAVCC                | MGTAVTT |                  |           |         |
| 1                      | 1       | 4.7              | 10%       | Ceramic |
| 2                      | 2       | 0.1              | 10%       | Ceramic |

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Sheet: /FPGA/FPGA Power Inputs/

File: FPGA\_PWR.kicad\_sch

Title: ThunderScope

Size: A4

Date:

KiCad E.D.A. 9.0.3

Rev: 5.3

Id: 11/20

Drawn by: Aleksa Bjelogrić

# FPGA Voltage Regulators

The recommended power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO  
1V,1V,1.8V,2.5V and 3.3V

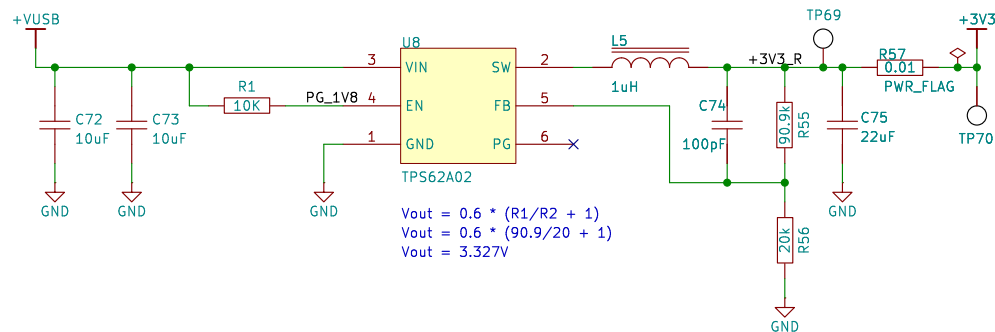
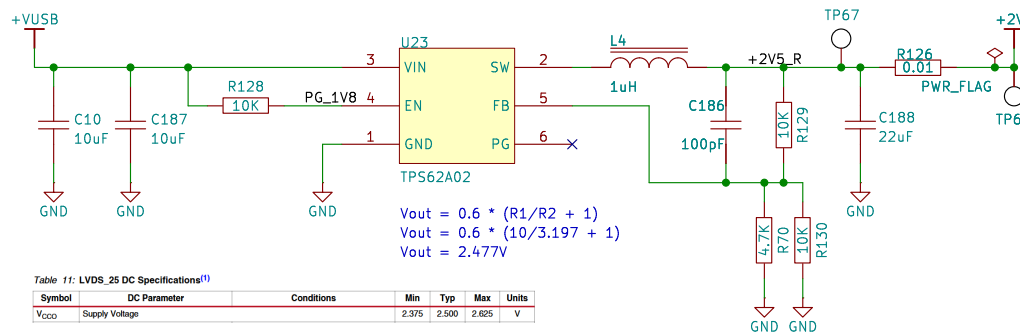
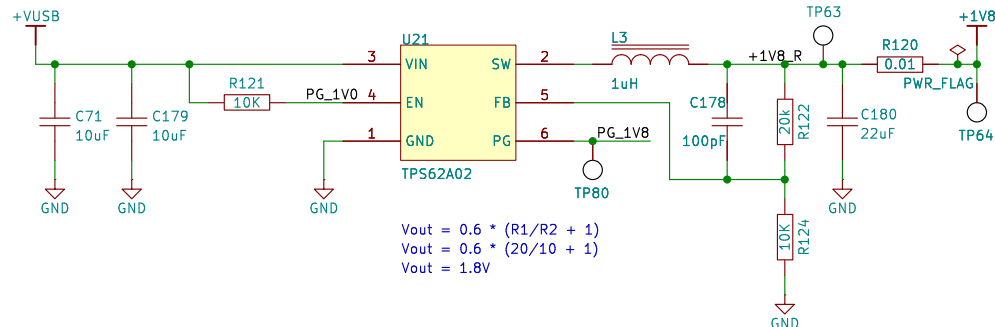
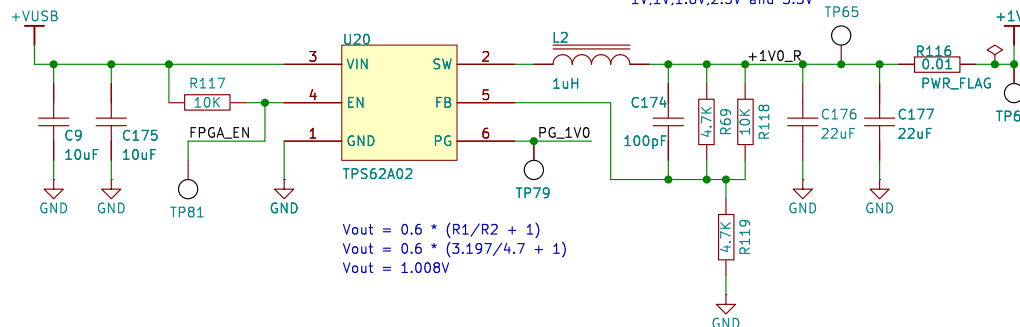
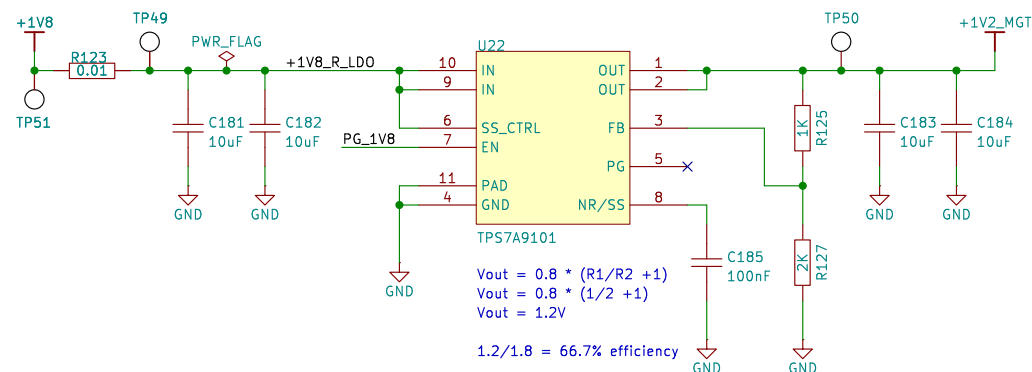


Table 11: LVDS\_25 DC Specifications<sup>(1)</sup>

| Symbol           | DC Parameter   | Conditions | Min   | Typ   | Max   | Units |
|------------------|----------------|------------|-------|-------|-------|-------|
| V <sub>CCO</sub> | Supply Voltage |            | 2.375 | 2.500 | 2.625 | V     |



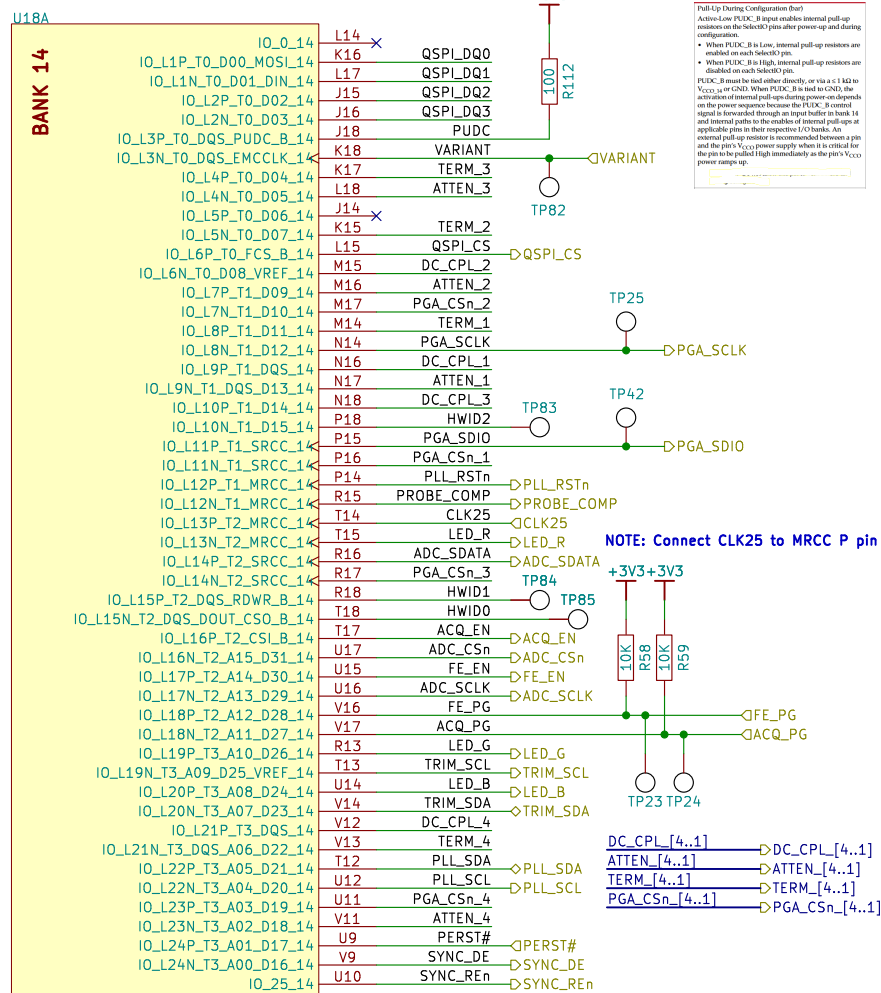
The recommended power-on sequence to achieve minimum current draw for the GTP transceivers is VCCINT, VMGTAVCC, VMGTAVTT  
1V,1V,1.2V

|                                                              |       |                            |           |
|--------------------------------------------------------------|-------|----------------------------|-----------|
| EEVengers                                                    |       | Drawn by: Aleksa Bjelogrić |           |
| Sheet: /FPGA/FPGA Voltage Regs/<br>File: FPGA_VREG.kicad_sch |       |                            |           |
| Title: ThunderScope                                          |       |                            |           |
| Size: A4                                                     | Date: |                            | Rev: 5.3  |
| KiCad E.D.A. 9.0.3                                           |       |                            | Id: 12/20 |



# FPGA IO Banks

Full TP Coverage on Bank 14 Signals (except PUDC)



FPGA Configuration

A High signal on the DONE pin indicates completion of the configuration sequence. The DONE output is an open-drain output by default.

**Note:** DONE has an internal pull-up resistor of approximately 10 kΩ. There is no setup/hold requirement for the DONE register. These changes, along with the DonePipe register software default, eliminate the need for the DriveDONE driver-option. External 330Ω resistor circuits are not required but can be used as they have been in previous generations.

Connect INIT\_B to a ≤ 4.7 kΩ pull-up resistor to V<sub>CC0\_0</sub> to ensure clean Low-to-High transitions.

Connect PROGRAM\_B to an external ≤ 4.7 kΩ pull-up resistor to V<sub>CC0\_0</sub> to ensure a stable High input, and

Table 2-1: 7 Series FPGA Configuration Modes

| Configuration Mode | M[2:0] | Bus Width  | CCLK Direction |
|--------------------|--------|------------|----------------|
| Master SPI         | 001    | x1, x2, x4 | Output         |

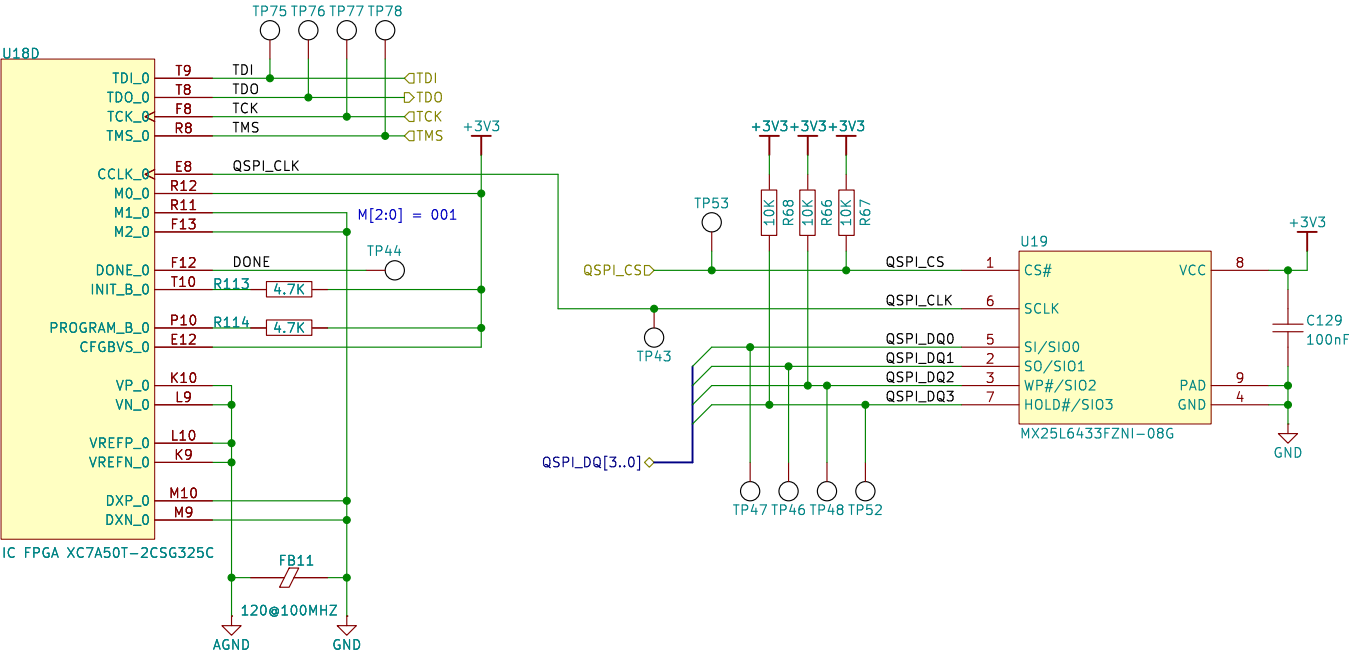
Table 2-6: Spartan-7, Artix-7 and Kintex-7 FPGA Configuration Mode, Compatible Voltages, and CFGBVS Connection

| Configuration Mode        | Banks Used | Configuration Interface I/O Voltage | HB Bank 0 V <sub>CC0_0</sub> | HB Bank 14 V <sub>CC0_14</sub> | HB Bank 15 V <sub>CC0_15</sub> | CFGBVS             |
|---------------------------|------------|-------------------------------------|------------------------------|--------------------------------|--------------------------------|--------------------|
| JTAG (only)               | 0          | 3.3V                                | 3.3V                         | Any                            | Any                            | V <sub>CC0_0</sub> |
|                           |            | 2.5V                                | 2.5V                         | Any                            | Any                            | V <sub>CC0_0</sub> |
|                           |            | 1.8V                                | 1.8V                         | Any                            | Any                            | GND                |
|                           |            | 1.5V                                | 1.5V                         | Any                            | Any                            | GND                |
|                           |            | 3.3V                                | 3.3V                         | 3.3V                           | Any                            | V <sub>CC0_0</sub> |
| Serial, SPI, or SelectMAP | 0, 14†1    | 2.5V                                | 2.5V                         | 2.5V                           | Any                            | V <sub>CC0_0</sub> |
|                           |            | 1.8V                                | 1.8V                         | 1.8V                           | Any                            | GND                |
|                           |            | 1.5V                                | 1.5V                         | 1.5V                           | Any                            | GND                |
|                           |            | 3.3V                                | 3.3V                         | 3.3V                           | 3.3V                           | V <sub>CC0_0</sub> |
|                           |            | 2.5V                                | 2.5V                         | 2.5V                           | 2.5V                           | V <sub>CC0_0</sub> |
| BPI†2                     | 0, 14, 15  | 1.8V                                | 1.8V                         | 1.8V                           | 1.8V                           | GND                |
|                           |            | 1.5V                                | 1.5V                         | 1.5V                           | 1.5V                           | GND                |

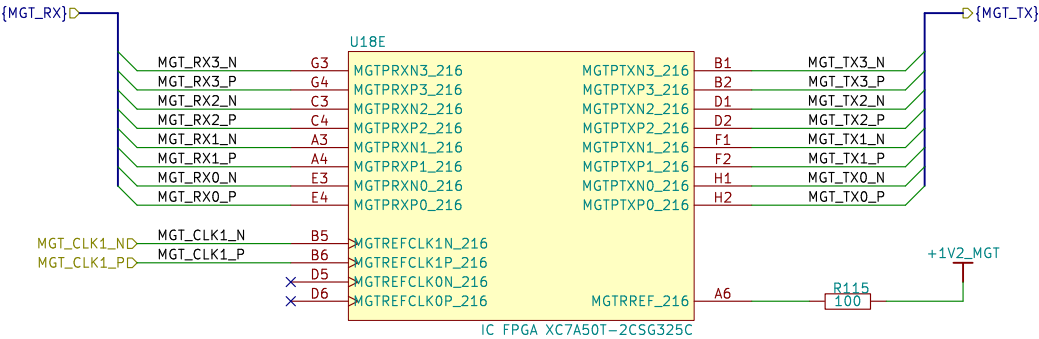
Notes:

1. BPI†2 for Multiboot or Fallback are in bank 15 but are typically only used in BPI mode and not supported in SPI mode.

2. BPI mode is not available in the Spartan-7 family.



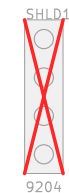
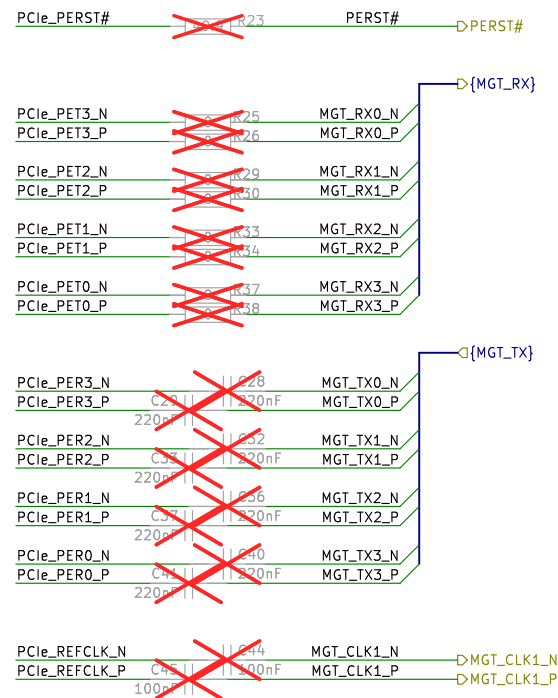
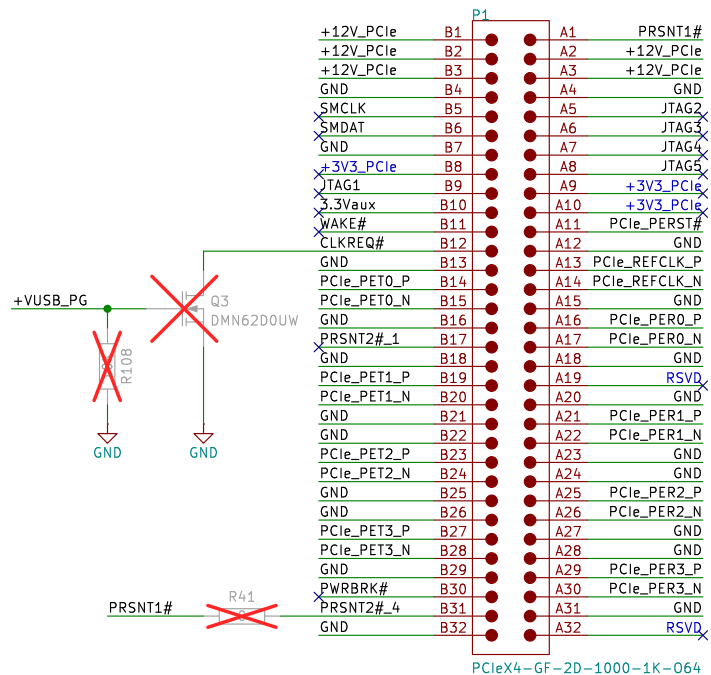
FPGA Transceivers



## PCIe x4 Edge Connector

AC Couple PER Lines with 220nF  
AC Couple REFCLK Lines with 100nF

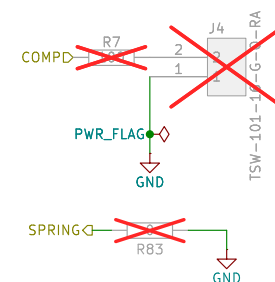
## PCIe bracket



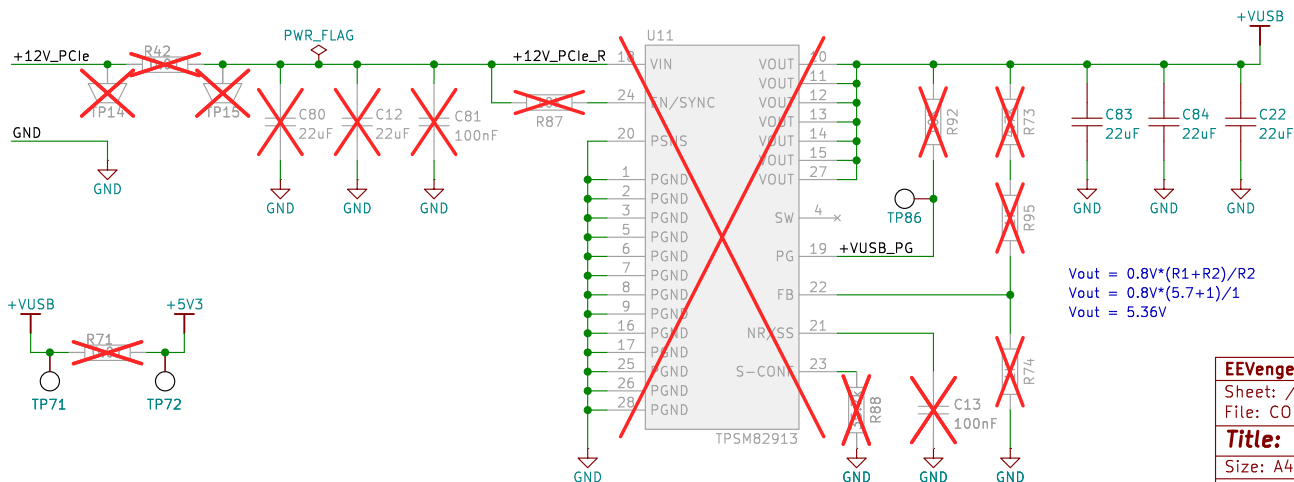
## Variant Pin Strap



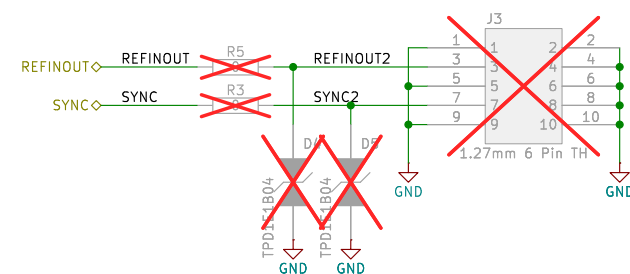
## Probe Comp



## 5V3 Buck Converter



## Refclock and Sync Header



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Sheet: /TS-PCIe Components/  
File: CON\_PCl\_e\_X4.kicad\_sch

Title: ThunderScope

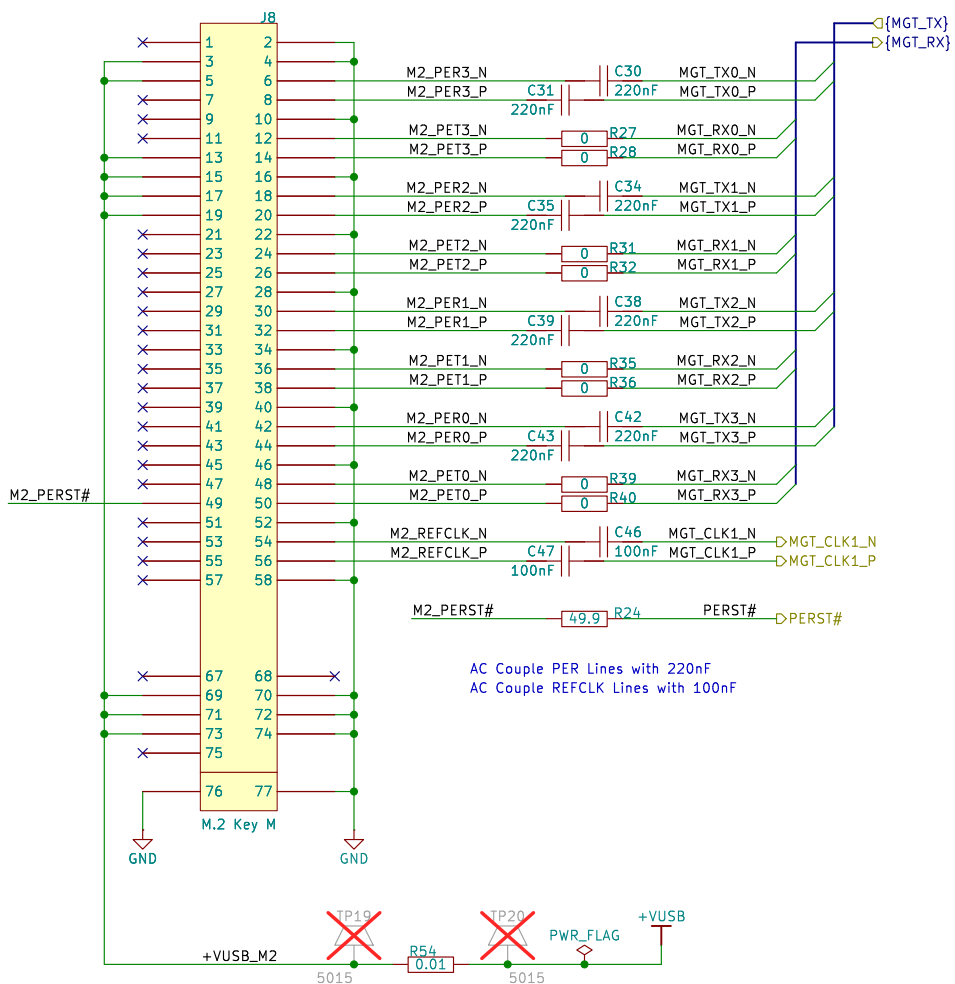
Size: A4 Date:

Rev: 5.3

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Id: 13/20

## M.2 Key M Connector – Custom Pinout

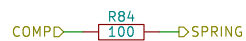


NOTE: The TB/USB4 adaptor must be modified to give us VUSB instead of 3V3

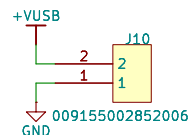
## Interposer Standoffs



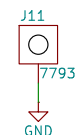
## Probe Comp



## Fan Connector



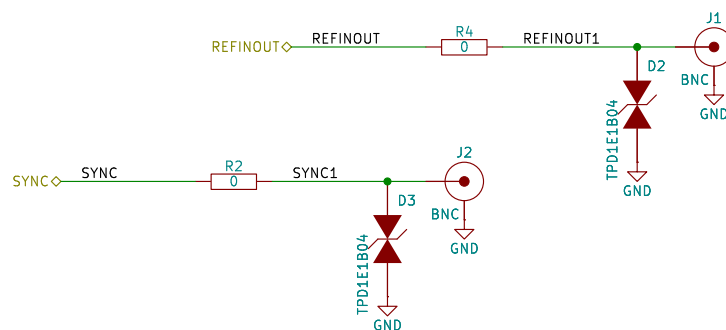
## Ground Lug



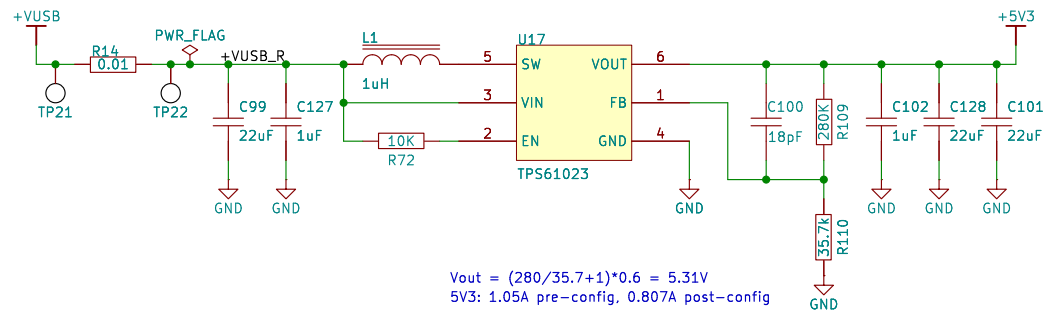
## Variant Pin Strap



## Refclock and Sync BNCs



## 5V3 Boost Converter



|                                                         |       |                            |           |
|---------------------------------------------------------|-------|----------------------------|-----------|
| EEVengers                                               |       | Drawn by: Aleksa Bjelogrić |           |
| Sheet: /TS-USB4 Components/<br>File: M2_KEY_M.kicad_sch |       |                            |           |
| Title: ThunderScope                                     |       |                            |           |
| Size: A4                                                | Date: |                            | Rev: 5.3  |
| KiCad E.D.A. 9.0.3                                      |       |                            | Id: 14/20 |